

Electronic diaphragm leakage sensor

Installation and operating instructions, supplement

Original installation and operating instructions.

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Warning

These complete installation and operating instructions are also available on www.grundfos.com.



Prior to installation, read these installation and operating instructions. Installation and operation must comply with local regulations and accepted codes of good practice.

In addition to these instructions, observe the general installation and operating instructions supplied with the pump.

1. General information

Pumps with diaphragm leakage detection (MBS) have a special dosing head flange for an optoelectronic sensor. The pump is supplied with the diaphragm leakage sensor (MBS) already installed.

1.1 Warranty

A guarantee claim in accordance with our general terms of sale and delivery is only valid if the following requirements are fulfilled:

- The product is used in accordance with the information within this manual.
- The product is not dismantled or incorrectly handled.
- The maintenance is carried out by authorised and qualified personnel.

1.2 Identification of safety instructions in this manual

If the safety instructions or other advice in this manual are not observed, it may result in personal injury or malfunction and damage to the pump. The safety instructions and other advice are identified by the following symbols:

**Warning**

If these safety instructions are not observed, it may result in personal injury!

Caution

If these safety instructions are not observed, it may result in malfunction or damage to the equipment!

Note

Notes or instructions that make the job easier and ensure safe operation.

2. Technical data

Product number **96725701**:

Model 230 V (+ 10 %/- 10 %)

Product number **91835310**:

Model 115 V (+ 10 %/- 10 %)

- Contact load: 250 V / 6 A, max. 550 VA
- Power consumption: 1.15 VA
- Enclosure class: IP65
- Permissible temperature range: 0 °C to +40 °C.

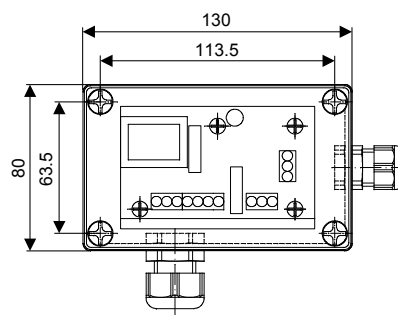
2.1 Dimensional sketch (electronics enclosure)

Fig. 1 Electronics enclosure

3. Function

Pumps prepared for diaphragm leakage detection:

- Special dosing head flange for inserting the optoelectronic sensor
- The optoelectronic sensor contains
 - infrared sender
 - infrared receiver.

In case of a leaking diaphragm

- The dosing liquid penetrates the dosing head flange.
 - The light refraction will be changed.
- The sensor produces a signal.
 - The electronics switches two contacts. These contacts can for instance be used to trigger an alarm device or to switch off the pump.

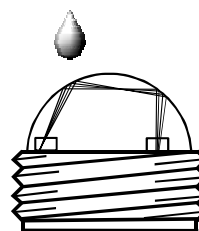


Fig. 2 Diaphragm leakage sensor

TM03 6381 4506

TM03 6382 4506

4. Electrical connection of the electronics

Warning

Electrical connections must only be carried out by qualified personnel!

Disconnect the power supply before connecting the power supply cable and the relay contacts! Observe the local safety regulations!

Protect the cable connections and plugs from corrosion and moisture.

Before connecting the power supply cable, check that the supply voltage stated on the pump nameplate corresponds to the local electricity supply. An incorrect power supply could destroy the unit!

Caution

To ensure electromagnetic compatibility (EMC), the input cables and current output cables must be screened.

1. Connect the screen at one end to PE.
– Refer to the connection diagram!
2. Route input cables, current output cables and power supply cables in separate ducts.
3. Connect the device to the power supply according to the connection diagram.
4. Connect the electronics with the sensor according to the connection diagram.

Warning

The potential-loaded contact 1, terminals 6 and 7, is loaded with supply voltage.

Switch off the power supply before connecting contact 1!

The contacts have no protective circuits.

Only pure ohmic loads may be switched.

For switching the pump motor, a contactor has to be connected inbetween.

Caution

5. Connect contacts 1 and 2 according to individual needs. Observe the general installation and operating instructions supplied with the pump, see section Electrical connections.

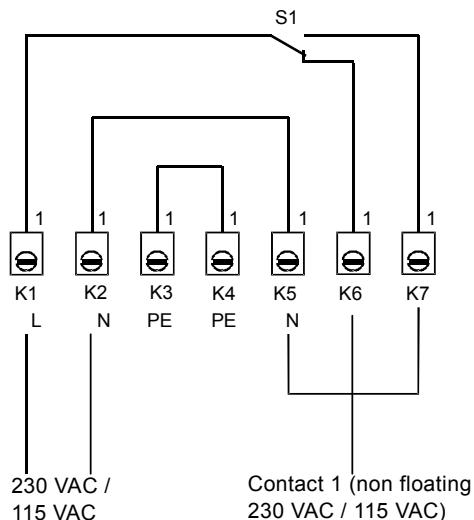


Fig. 4 Electrical connection of the electronics

4.2 Screwing the sensor into the dosing head

- Screw the sensor from the lower side into the hole of the dosing head flange (M14 x 1.5).
– Now the diaphragm leakage sensor is ready for start-up.

4.1 Relay outputs

Note

The relay output connection depends on the application and the connected actuators.

- Interference suppression is required for inductive loads (also relays and contactors).
- If this is not possible, protect the relay contacts using a suppressor circuit as described below.

With AC voltage

Current up to	Capacitor C	Resistor R
60 mA	10 μ F, 275 V	390 Ω , 2 W
70 mA	47 μ F, 275 V	22 Ω , 2 W
150 mA	100 μ F, 275 V	47 Ω , 2 W
1.0 A	220 μ F, 275 V	47 Ω , 2 W

With DC voltage

- Connect the free-wheeling diode parallel to the relay or contactor.

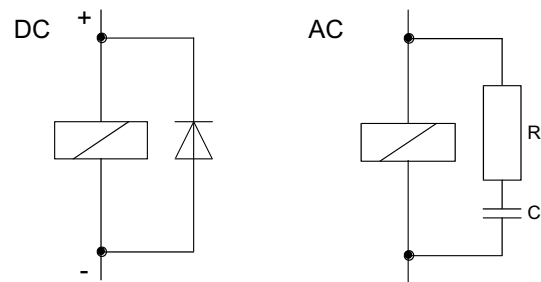


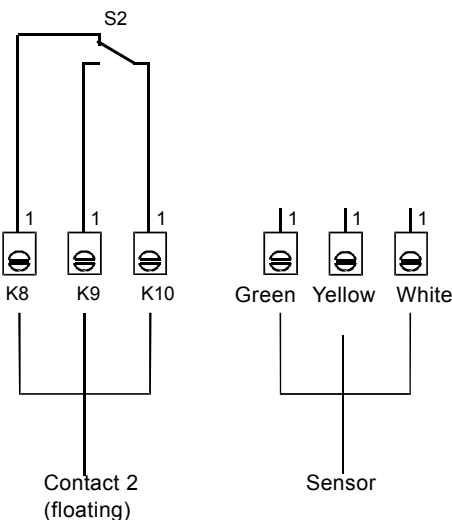
Fig. 3 Suppressor circuit DC/AC

Caution

Provide relay outputs on site with an appropriate back-up fuse!

Note

These connections depend on the type of actuator used and should only be understood as guidelines. Refer to actuator documentation.



TM03 7209 4513

TM03 6385 4506

5. Start-up

Caution *Carry out a functional check before start-up!*

Functional check

- Dip the sensor into water.
 - Green and red LEDs are on:
Sensor and electronics are ready for operation!
 - One or more LEDs are off:
Sensor or electronics is defective!
Call Grundfos service.
- Carefully dry the sensor.
 - Only the green LED is still on:
Sensor and electronics are ready for operation!
 - The red LED is still on:
Sensor or electronics is defective!
Call Grundfos service.



Warning

Do not open the electronics or sensor!

Repairs must only be carried out by authorised and qualified personnel!

5.1 Using the contacts

- Terminals 6 and 7 (potential-loaded)
 - for instance for switching off the pump in case of a diaphragm leakage.
- Terminals 8, 9 and 10 (potential-free)
 - for instance for triggering an alarm device.

5.2 Description of the device

There are a green and a red light-emitting diode (LED) at the electronics.

- Green LED
 - shows that the system is ready for operation.
 - The LED is only on when the sensor is connected to the electronics.
If the LED is off in this case, either the sensor or the cable is defective or wrongly connected.
- Red LED
 - shows that a diaphragm leakage has been detected.
 - The green LED is still on.

6. Maintenance



Warning

Do not open the electronics or sensor!

Repairs must only be carried out by authorised and qualified personnel!

6.1 Sensor

- Clean the sensor in case of malfunction.
- If the sensor still does not operate correctly, replace it.
 - Product number 96609034:
Optoelectronic sensor with 3 metres cable.

6.2 Electronics

- No maintenance is possible by the user.
- If the electronics does not operate correctly, call Grundfos service.

7. Disposal

This product or parts of it must be disposed of in an environmentally sound way. Use appropriate waste collection services. If this is not possible, contact the nearest Grundfos company or service workshop.

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